

Graph Signal Processing: Towards The Diffused Spectral Clustering

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Abstract—Graph signal processing [7] is an emerging field of research. When the structure of signals can be represented as a graph, it allows to fully exploit their inherent structure. It has been shown that the normalized graph Laplacian matrix plays a major role in the characterization of a signal on a graph. Moreover, this matrix plays a major role in clustering large data set. In this paper, we present a novel handwritten digits [2] clustering algorithm based on the normalized graph laplacian properties. This method is very efficient and can easily defeat the other algorithm of the state of art. We reach a spectacular error rate of the order of 1%. This emphasize the advantages of introducing the graph signal processing domain in the data science field.

Index Terms—Graph signal processing, Spectral clustering, Diffusion, Normalized Graph Laplacian.

I. INTRODUCTION

During the recent years, the analysis and processing of large-scale datasets [3] using graphs has become very useful. In fact, many kinds of data domains such as social and economic networks, electric grids, neuronal networks and images databases [3] require a graph representation of their structure. Each of these structures usually carries out information that flow between different elements of the network. For example, in a neural network, a neuron is activated after receiving an electric excitation, and the activation of a neuron usually influences the nearby neurons. In the case of economic networks, we can consider the economic crisis as a flow that spreads from one bank to another. This need to represent these phenomena has lead to the development of a new field: the graph signal processing. Indeed, a continuous signal can be sampled according to a specific frequency and the sampled discrete signal that is obtained is usually carried out on a graph [7]. By this way, we obtain at the same time a representation of the structure of the network as well as of the information flowing through it. For instance, a sound signal can be represented on a linear or a ring graph. However, a picture is usually represented on a grid graph where each pixel is linked to its four or eight nearest neighbors [7]. Weighted graphs are particularly used to represent the links and similarities between the different elements of a network. The advantage about signals on graphs is the fact that they can be processed in a way analogous to the classical signal processing.

One of the main applications of graph signal processing [7] today appear in the field of artificial intelligence and especially in machine learning. Partitionning a large data

set into homogeneous clusters is among the most known applications in this domain. The litterature [3] gives us many clustering algorithms but the graph signal processing using the spectral clustering method seems to be more efficient.

In this paper, we present a novel method of clustering handwritten digits taken from the MNIST database [2] based on the graph signal processing. It's a combination between a graph feature space transformation of the digits and the recent clustering algorithm known as spectral clustering [8].

The remainder of the paper is as follow. In the next section, we provide some background from the graph signal processing domain [8]. In section III, we present the graph feature space transformation. Then section IV introduces our novel method called diffused spectral clustering. Then, we discuss our results compared to the state of art. Section VII concludes the paper.

II. GRAPH SIGNAL PROCESSING

Let us introduce notations first. We consider a weighted, simple, undirected graph $G = (E, V, W)$ where E represents the set of edges and V the set of vertices. Without a loss of generality, we consider V to be the set of integers between 1 and $N = |E|$. We equip G with a $N \times N$ adjacency matrix W defined as follows :

$$W_{i,j} \triangleq \begin{cases} \text{The weight of the edge connecting } i \text{ and } j \\ 0 \text{ if no such edge exists} \end{cases} \quad (1)$$

When the edge weights are not naturally defined by an application, one common way to define the weight of an edge connecting vertices i and j is via a similarity function [7] like a distance:

$$W_{i,j} = \text{dist}(i, j) \quad (2)$$

Where $\text{dist}(i, j)$ may represents a physical distance between two feature vectors describing the nodes i and j .

We also define the $N \times N$ diagonal degree matrix D as:

$$D_{i,i} = d_i = \sum_{k=0}^N W_{i,k} \quad (3)$$

For instance, a social network can be represented by a weighted, simple, undirected graph, where the vertices are the individuals and the edges represent the friendship bond between two individuals. In this case, the degree matrix gives us an idea about how important are the friendship links of each individual.

We then introduce the non-normalized graph Laplacian L [6]:

$$L \triangleq D - W$$

This matrix turns to have a major importance as it stands for a differentiation operator for a signal over a graph.

We remind that a signal over a graph G is a vector $x \in \mathbb{R}^N$ where the i^{th} component of the vector x represents the function value at the i^{th} vertex of V . The Laplacian's i^{th} component of such a signal is the vector [6]:

$$(Lx)(i) = \sum_{j=1}^N W_{i,j}[x(i) - x(j)] \quad (4)$$

For example in the case of the social network, a signal can represent a rumor: the individuals who received the rumor are given the value 1 and those who did not are given the value 0. We obtain therefore a binary signal on graph.

When working with L_2 -norm, it makes sense to use instead the normalized graph Laplacian, defined as [6]:

$$\mathcal{L} = D^{-\frac{1}{2}} \cdot L \cdot D^{-\frac{1}{2}} \quad (5)$$

Since the normalized (or standard) graph Laplacian is a real valued symmetric matrix, it can be diagonalized using an orthonormal basis. We denote a corresponding set of orthonormal eigenvectors by $\{\mu_l\}_{l=1,2,\dots,N}$ and the set of associated real, non-negative eigenvalues by $\{\lambda_l\}_{l=1,2,\dots,N}$ when those are ordered from the lowest eigenvalue to the largest one.

In particular, we have [6]:

$$\mathcal{L}\mu_l = \lambda_l\mu_l \quad (6)$$

It is well-known that [7]:

$$0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N \triangleq \lambda_{max} \leq 2 \quad (7)$$

The literature gives many results binding eigenvalues with properties of the graph. As an example, the number of connected components of the graph is given by the multiplicity of the eigenvalue zero. For instance, if the graph is connected, the multiplicity of the eigenvalue zero is one. Also the highest eigenvalue is equal to 2 if and only if the graph is bipartite.

The first eigenvector μ_1 has a closed-form given by the following formula:

$$\mu_1(i) = \sqrt{\frac{d_i}{\sum_{u \in V} d_u}} \quad (8)$$

In the case of regular graphs, all the vertices have the same degree so μ_1 is a constant vector.

Eigenvectors of the graph normalized Laplacian extend the principles of the Fourier transform for classical signal processing. To understand this bindings, let us recall that the classical Fourier transform of a signal f is given by [7]:

$$\tilde{f}(\omega) = \int_{\mathbb{R}} f(x) \cdot e^{-i\omega x} dx \quad (9)$$

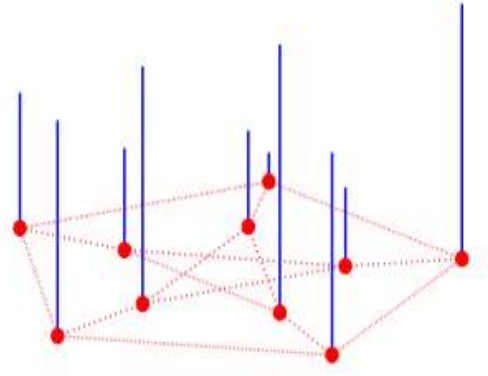


Figure 1. A positive graph signal defined on the Peterson graph. The height of each blue bar represents the signal value at the vertex where the bar originates [7].

Having:

$$\frac{d^2}{dx^2} e^{-i\omega x} = -\omega^2 e^{-i\omega x} \quad (10)$$

We can notice that $e^{-i\omega x}$ is the eigenvector of the Laplace operator $\frac{d^2}{dx^2}$ associated with the eigenvalue $-\omega^2$. On the other hand, we have [7]:

$$\mathcal{L}\mu_l = \lambda_l\mu_l \quad (11)$$

so the frequencies in classical signal processing are analogous to the eigenvalues of the normalized Laplacian in graph signal processing. Consequently the Fourier transform $\hat{x}$ of a signal x on graph G is defined as [7]:

$$\hat{x}(\lambda_l) = \sum_{i=1}^N x(i) \mu_l^*(i) \quad (12)$$

Where μ_l^* represents the complex conjugate of the eigenvector μ_l .

And the inverse graph Fourier transform is defined as [6]:

$$x(i) = \sum_{l=1}^N \hat{x}(\lambda_l) \mu_l(i) \quad (13)$$

Finally, to characterize the smoothness of a signal on a graph G , one can use the Dirichlet form [6]:

$$S(x) = \frac{x^T \mathcal{L} x}{\|x\|^2} = \frac{1}{\|x\|^2} \sum_{l=1}^N \lambda_l (\langle x, \mu_l \rangle)^2 \quad (14)$$

The smaller is $S(x)$, the smoother is the signal x .

III. GRAPH FEATURE SPACE TRANSFORMATION

The objective of this section is to well define a suitable metric in order to compare signals seen as vectors defined on the vertices of a given graph.

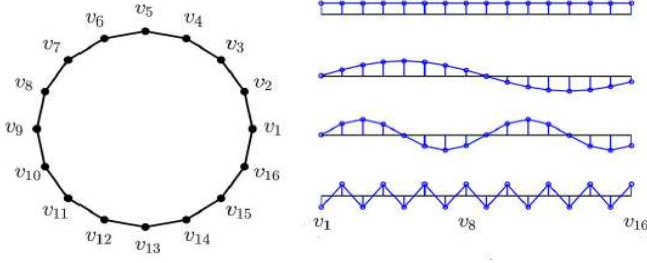


Figure 2. Representation of the 16 cycle graph Laplacian eigenvectors. The eigenvectors exhibit the sinusoidal characteristics of the Fourier Transform basis. Signals defined on this graph are equivalent to classical discrete, periodic signals [7].

A. Metrics

A metric $d : E \times E \rightarrow \mathbb{R}_+$ is a function from pairs of elements in the space E to a positive real value satisfying the following three conditions for every $u, v, w \in E$:

- Symmetry: $d(u, v) = d(v, u)$
- Identity: $d(u, v) = 0 \Leftrightarrow u = v$
- Triangular inequality: $d(u, v) \leq d(u, w) + d(w, v)$

B. Norms

A closely related definition of a metric is that of a norm, which is a function $\|\cdot\| : E \rightarrow \mathbb{R}_+$ satisfying for every vectors $u, v \in X$ and every scalar $\lambda \in \mathbb{R}$:

- Positiveness: $\|u\| \geq 0$ and $\|u\| = 0 \Leftrightarrow u = 0$
- Positive homogeneity: $\|\lambda u\| = |\lambda| \|u\|$
- Subadditivity: $\|u + v\| \leq \|u\| + \|v\|$

We can easily verify that the function $d : E \times E \rightarrow \mathbb{R}$ defined as $d(u, v) := \|u - v\|$ is a metric for every $u, v \in E$

C. Heat Diffusion On Graph

Given an arbitrary graph $G = (V, E, W)$, its normalized graph Laplacian $\mathcal{L}$, a signal x_0 defined on G and seen as a vector $x_0 = [x_{01}, \dots, x_{0n}] \in \mathbb{R}^n$ and a nonnegative constant α , we define the time-varying signal $x \in \mathbb{R}^n$ as the solution of the following differential equation [8]:

$$\begin{cases} x(0) = x_0 \\ \frac{\partial x}{\partial t} = -\alpha \mathcal{L} x \end{cases} \quad (15)$$

This differential equation represents the heat diffusion on the graph G in so far as $-\mathcal{L}$ can be seen as the analogous of the well-know continuous Laplacian operator used in the classical heat diffusion description. The signal x describes the temperature distribution at a given time while x_0 describes the initial distribution before any diffusion. The positive constant α represents the thermal conductivity controlling the heat diffusion rate. The greater is α the faster will be x smooth. We can easily verify that the solution of is given by the following formula [8]:

$$x(t) = e^{-\alpha \mathcal{L} t} x_0 \quad (16)$$

This formula allows us to compute the temperature distribution at any time point with only the knowledge of the initial

distribution x_0 and the normalised graph laplacian $\mathcal{L}$ of the structure carrying out the signal x . One can notice that $x(t)$ tends to an uniform distribution since time grows.

D. Diffusion Distance

With the same notations as above, we introduce the diffusion distance $d^{\mathcal{L}}(u, v)$ between the signals u and v defined as [8]:

$$d^{\mathcal{L}}(u, v) := \left\| \int_0^{+\infty} e^{-t} e^{-\alpha \mathcal{L} t} (u - v) dt \right\| \quad (17)$$

for a given nonnegative scalar $\alpha > 0$ which corresponds to the thermal conductivity in (15). One can notice that the diffusion distance incorporates the structure of the graph which carry out the signals u and v . By this way, we can well-determine the proximity between u and v instead of using classical distances as the L_2 -distance which doesn't take into account the data type. Noticing that the primitive of the matrix $e^{-t} e^{-\alpha \mathcal{L} t} = e^{-(I + \alpha \mathcal{L})t}$ is $-(I + \alpha \mathcal{L})^{-1} e^{-(I + \alpha \mathcal{L})t}$, we can easily verify that [8]:

$$d^{\mathcal{L}}(u, v) = \left\| (I + \alpha \mathcal{L})^{-1} (u - v) \right\| \quad (18)$$

Thus, for a input vector $u \in \mathbb{R}^n$ and a nonnegative scalar $\alpha > 0$, we define the diffusion distance transformation $u_{\mathcal{L}}$ of u as:

$$u_{\mathcal{L}} = (I + \alpha \mathcal{L})^{-1} u \quad (19)$$

By this way, the diffusion can be interpreted as a feature space transformation on the assumption that there exist relations between the features. This relations can easily be interpreted as a graph describing the structure likely to carry out the features of any given signal.

For instance, let us consider the MNIST handwritten digit database [2]. Each picture of this database consists of a gray scaled image of size 28×28 pixels representing a given handwritten digit. Therefore, we can see each observation in the MNIST database [2] as a signal $x \in \mathbb{R}^{784}$ where the value of each of its components corresponds to the associated pixel intensity. Nonetheless, there are relations between the 784 features imposed by the underlying structure of an image. Indeed, we construct a undelying graph where each pixel is linked by an unit weight edge to its four (4NN) or eight (8NN) nearest neighbours. By denoting $\mathcal{L}$ the normalized graph Laplacian of the underlying graph built, we are allowed to use the formula (19) in order to obtain the transformed versions by diffusion distance of each handwritten digit. Figure 3 shows us an observation of the digits 1 and 2 drawn from MNIST [2], and their transformed versions by the 4NN and 8NN underlying graph, we choose $\alpha = 10$.

From figure 4, it is clear that the feature transformation make easy the comparison between the digits. The distance between the transformed versions of two similar digits tends to be lesser.

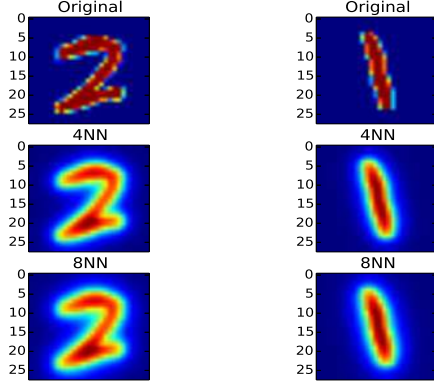


Figure 3. Observations of the digits 1 and 2, and their transformed smoothed versions using the 4NN and 8NN underlying graph. The transformed versions of the digits facilitate the comparison between the original digits which can make easy the partitionning of them.

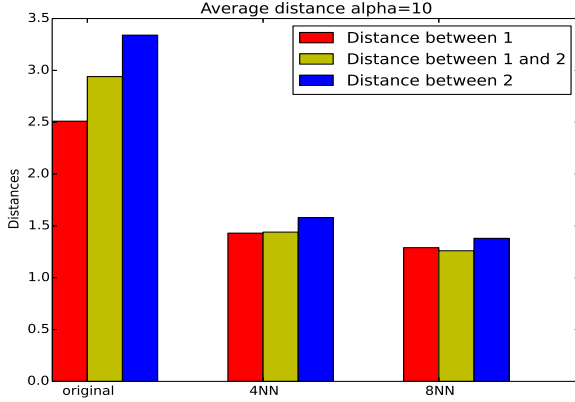


Figure 4. Average distances between digits before and after applying a feature transformation considering the 4NN or the 8NN underlying graph.

IV. DIFFUSED SPECTRAL CLUSTERING

A. Normalized Spectral Clustering

In this section, we briefly recall the principle of the spectral clustering algorithm [8] applied in order to partition a graph into several homogeneous subgraphs [9]. For this purpose, let us consider a unweighted and undirected graph $G(V, E, A)$ where the vertex set $V = \{v_i\}_{i \in 1..n}$ of n vertices represents the database to cluster. We build the weight matrix W of size $n \times n$ on the model of Gaussian kernel as follow:

$$W_{i,j} = \exp(-\beta \|v_i - v_j\|_2^2) \quad (20)$$

where v_i represents the vertex at node i , and $\|\cdot\|_2$ represents the classical euclidean norm.

Thus, we build the $n \times n$ unweighted adjacency matrix A on the model of nearest neighbours (one gives a γ threshold) as follow:

Algorithm 1 Normalized Spectral Clustering [8]

- **Input :**
 - W : $n \times n$ weighted adjacency matrix of G
 - k : Number of clusters expected
 - Compute the degree matrix D and the normalized graph laplacian $\mathcal{L}$
 - Let $U \in \mathbb{R}^{n \times k}$ be the solution of containing the first k eigenvectors $\mu_1, \dots, \mu_k$ of the matrix $\mathcal{L}$ as columns.
 - Normalize each row of U to get U_{norm} .
 - Let $y_i \in \mathbb{R}^k$ be the i^{th} row of U_{norm} .
 - Launch the k -means algorithm to partition the $y_i \in \mathbb{R}^k$ into k clusters $C_1, \dots, C_k$.
 - **Output :**
 - $C_1, \dots, C_k$: the clusters expected
-

$$A_{i,j} = \begin{cases} 1 & \text{if the node } i \text{ is a node } j \text{ neighbor} \Leftrightarrow W_{i,j} > \gamma \\ 0 & \text{otherwise} \end{cases} \quad (21)$$

We recall that the normalized graph Laplacian associated to the graph G is computed with regards to the following formula:

$$\mathcal{L} = D^{-\frac{1}{2}}(D - A)D^{-\frac{1}{2}} \quad (22)$$

where D is the degree matrix associated to the graph G .

We are now looking at partitionning the vertices v_i of G into k different subsets called clusters, such that the vertices belonging to the same cluster are connected by edges of large weights which can be considered as a similarity measure. This problem can be solved using the algorithm proposed in [4] which solves the following optimization problem :

$$\min_{U \in \mathbb{R}^{n \times k}} \text{tr}(U^T \mathcal{L} U), \quad \text{such that } U^T U = I \quad (23)$$

Using the Rayleigh-Ritz theorem [4], we can show that the solution of the optimization problem is the matrix containing the first k eigenvectors associated to the k smallest eigenvectors of $\mathcal{L}$. The clustering is then computed by applying the k -means [3] algorithm to the row vector of the matrix U . So, the spectral clustering reduces the dimensionality [?] of the data before applying the k -means algorithm. This is generally done in order to avoid the curse of dimensionality [?] which foils any partitionning algorithm based on geometric approaches like the k -means algorithm.

We summarize the spectral clustering algorithm in algorithm 1.

B. Diffused Spectral Clustering

We recall that the goal of this paper is to introduce an efficient algorithm in order to well-partition a handwritten digit database taken from MNIST [2]. The idea is to combine the graph feature space transformation explained earlier with the normalized spectral clustering principle. Indeed, the graph feature space transformation is used in order to reduce the

Algorithm 2 Diffused Spectral Clustering

- **Input :**
 - The set $\{I_i\}$ of images drawn from MNIST. $I_i \in \mathbb{R}^{28 \times 28}$. Assume that $\text{card}(\{I_i\}) = N$
 - α the parameter of diffusion. Typically, $\alpha = 10$.
 - $\mathcal{L}$ be the normalized graph laplacian of an $4NN$ or $8NN$ underlying graph.
 - k : Number of clusters expected.
 - Transform each matrix $I_i \in \mathbb{R}^{28 \times 28}$ into a vector row $x_i \in \mathbb{R}^{1 \times 784}$.
 - Transform each x_i into $x_i^{\mathcal{L}} = (I + \alpha \mathcal{L})^{-1} x_i$ using the graph feature space transformation.
 - Let $G = (V, E, W)$ be the weighted complete graph of size N carrying out the set of transformed images $\{v_i = x_i^{\mathcal{L}}\}$. W is computed using equation 20.
 - Compute the unweighted adjacency matrix A using equation 21 and W .
 - Launch a normalized spectral clustering using A and k as input to get the clusters $C_1, \dots, C_k$
 - **Output :**
 - $C_1, \dots, C_k$: the clusters expected.
-

geometric distance (L_2 distance) between images describing the same digits. By this way, the k -means algorithm [3] which is based on geometric approaches will be efficiently applied to group the same digits and thus to cluster the input database.

The spectral clustering algorithm applied to the MNIST database is summarized in algorithm 2.

V. RESULT ANALYSIS

In this section, we are making some discussions about the diffused spectral clustering algorithm used in order to cluster samples taken from the MNIST database.

Without a loss of generality, we only deal with digits 1 and 2. The first step consists on choosing the best value of the parameter α . Figure 5 shows us that choosing a value of the parameter α of the order of 10 is a sufficient condition to reach the optimum error rate when we are dealing with the $4NN$ or $8NN$ underlying graph. Error rate means the percentage of wrong digits detected among the treated digits set. Naturally, the lesser is the error rate, the better is the clustering algorithm.

By setting the parameter $\alpha = 10$, we evaluate the error rate achieved by the diffused spectral clustering in both cases of the $4NN$ or $8NN$ underlying graph.

We are then interested in comparing the diffused spectral clustering performance to those of another clustering algorithm from the state of art like the k -means algorithm [3]. We recall that the only difference between k -means and spectral clustering is the dimensionality reduction applied by the last one on the original data. Figure 6 shows us this comparison.

From figure 6, we can conclude about the importance of the diffused spectral clustering. We reach a very low error rate (0.01) when we are dealing with the $4NN$ or $8NN$ underlying graph.

In order to emphasize this result, we compare it to the one obtained by computing a principal component analysis [1]

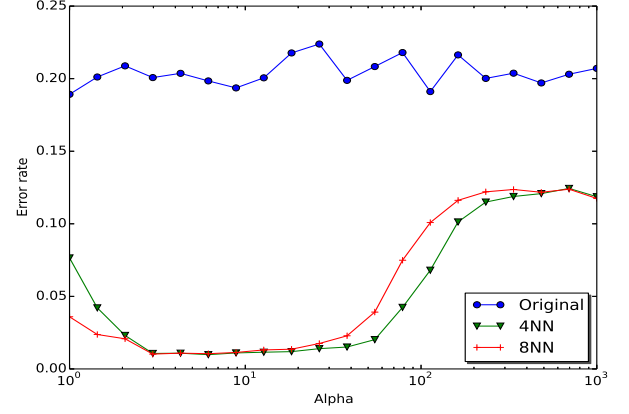


Figure 5. Error rate variations for 1000 samples of digit 1 and 2. We notice that the optimal value of the parameter α is of the order of 10.

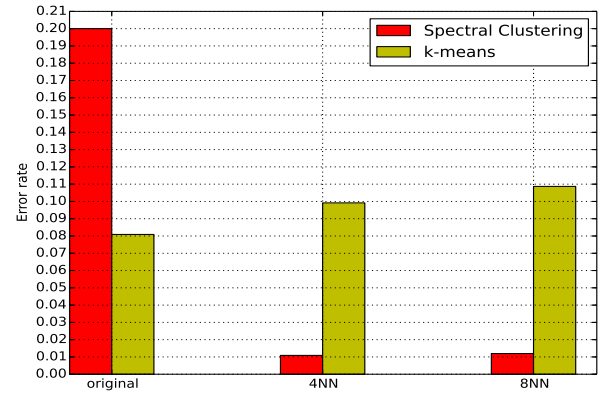


Figure 6. Clustering of 1000 samples of digit 1 and 2 using the spectral clustering vs k -means. We reach an optimal error rate when using the diffused spectral clustering with a diffusion parameter $\alpha = 10$.

(PCA) before launching the k -means. This has been chosen to well compare the dimensionality reduction influence on the clustering result. Figure 7 shows the result of this comparison.

So, by applying a graph feature space transformation (using the $4NN$ or $8NN$ underlying graph), the spectral clustering becomes a very efficient algorithm of data partitioning. What would be the result if the underlying graph chosen depends on the data values? We recall that the $4NN$ and $8NN$ underlying graph don't depend on the data values but only on the structure likely to carry out the data. One approach to construct an underlying graph depending on the data values is the covariance matrix. By this way, the adjacency matrix of the underlying graph will be the data covariance matrix. In this case, we have to find the best value of the diffusion parameter α which optimizes the error rate. From figure 8, we notice that the value of α has to be of the order of 3000, but it seems that this value is not very stable.

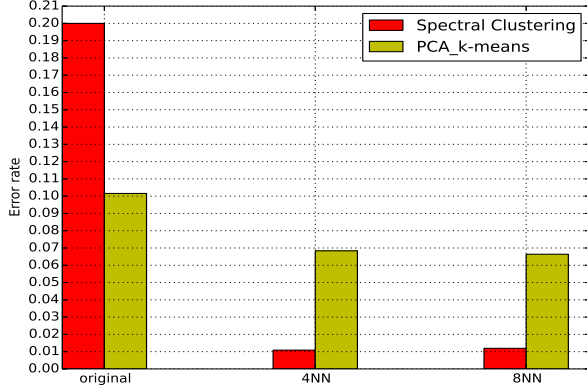


Figure 7. Clustering of 1000 samples of digit 1 and 2 using both spectral clustering and k -means after a PCA [1].

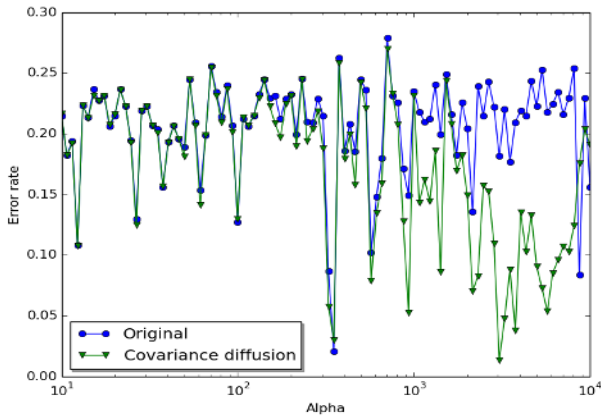


Figure 8. Error rate variations for 1000 samples of digit 1 and 2. We notice that the optimal value of the parameter α is of the order of 3000.

Comparison With The Anisotropic Diffusion

From figure 3, it is clear that the graph feature space transformation based on the heat diffusion principle gives us the smoothed version of any image. This transformation can be seen as an anisotropic diffusion which retains the image outlines. The literature gives us many method to achieve this. For instance, one can be interested in the Perona Malik anisotropic diffusion [5]. Figure 9 shows us the differences between our diffusion and the classical anisotropic diffusion operated by Perona-Malik [5].

We then compare the effect of such diffusion on clustering. For this purpose, we launch the spectral clustering on the Perona-Malik anisotropic diffused version of digits 1 and 2 and compare that with the diffused spectral clustering (4NN and 8NN underlying graph). Table I shows that even the Perona Malik diffusion [5] cannot compete with the underlying graph diffusion.

VI. FUTURE WORKS

Our future works consist on applying the diffused spectral clustering on a database other than MNIST [2] in order to

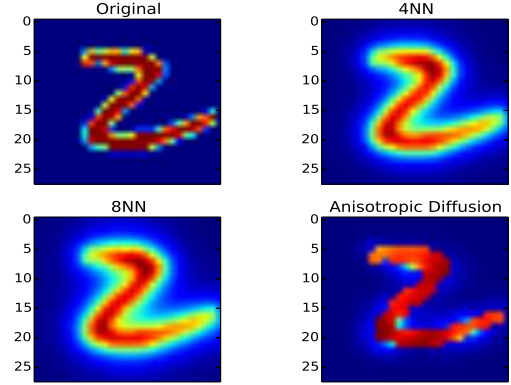


Figure 9. Visualisation of different diffusions applied to a digit 2 taken from MNIST.

Diffusion	4NN	8NN	Perrona-Malik
Error rate (%)	1.01	1.1	19.6

Table I

COMPARISON BETWEEN THE SPECTRAL CLUSTERING APPLIED ON THE 4NN, 8NN UNDERLYING GRAPH AND THE PERRONA-MALIK DIFFUSED VERSIONS OF 1000 DIGITS OF 1 AND 2 TAKEN FROM THE MNIST DATABASE.

partition data other than images. For example, we are thinking about adapting our method in order to cluster sound signals, or other complex signals carried out by complex structures like neural and social networks.

VII. CONCLUSION

This paper shows the importance of using the graph signal processing in applications such clustering. By this way, we have implemented a novel clustering algorithm for partitioning handwritten digits. It seems that it's very efficient and better than the state of art. Our algorithm is essentially based on the normalized graph Laplacian properties in terms of smoothness, and informations carried out by its eigenspace. These informations seem to be very useful to understand the implicit and explicit relationships in a database.

Therefore, the study of the normalized graph Laplacian spectrum provides us with solutions to some frequent applications. There are many other use cases that can be treated using the graph Laplacian method and that need to be considered in further studies.

REFERENCES

- [1] Xiaowen Dong, Dorina Thanou, Pascal Frossard, and Pierre Vandergheynst. Learning laplacian matrix in smooth graph signal representations. *arXiv preprint arXiv:1406.7842*, 2014.
- [2] Cheng-Lin Liu, Kazuki Nakashima, Hiroshi Sako, and Hiromichi Fujisawa. Handwritten digit recognition: benchmarking of state-of-the-art techniques. *Pattern Recognition*, 36(10):2271–2285, 2003.
- [3] Erwan Le Martelot and Chris Hankin. Fast multi-scale community detection based on local criteria within a multi-threaded algorithm. *arXiv preprint arXiv:1301.0955*, 2013.
- [4] Andrew Y Ng, Michael I Jordan, Yair Weiss, et al. On spectral clustering: Analysis and an algorithm. *Advances in neural information processing systems*, 2:849–856, 2002.

- [5] Pietro Perona, Takahiro Shiota, and Jitendra Malik. Anisotropic diffusion. In *Geometry-driven diffusion in computer vision*, pages 73–92. Springer, 1994.
- [6] Michael G Rabbat and Vincent Gripon. Towards a spectral characterization of signals supported on small-world networks. In *Acoustics, Speech and Signal Processing (ICASSP), 2014 IEEE International Conference on*, pages 4793–4797. IEEE, 2014.
- [7] David Shuman, Sunil K Narang, Pascal Frossard, Antonio Ortega, Pierre Vandergheynst, et al. The emerging field of signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular domains. *Signal Processing Magazine, IEEE*, 30(3):83–98, 2013.
- [8] DONG Xiaowen. *Multi-View Signal Processing and Learning on Graphs*. PhD thesis, ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE, 2014.
- [9] Yunpeng Zhao, Elizaveta Levina, and Ji Zhu. Community extraction for social networks. *Proceedings of the National Academy of Sciences*, 108(18):7321–7326, 2011.